

Project Workflow: Week 7

Phase: Scenery & Background

1. Objective

The goal of Week 7 was to enhance the game's visual immersion by implementing a comprehensive environment and dynamic background system. This included:

1. Implementing the Environment manager with Tree, Bush, Mountain, Lake, and Park classes.
2. Adding parallax scrolling so background layers move at different speeds to create a sense of depth.
3. Implementing animated Bird and Cloud elements alongside a dynamic dawn-to-dusk sky transition.

2. Detailed Task Breakdown

Implementing Environment Classes (Tree, Bush, Mountain, Lake, Park) Created a modular structure for background elements. Static and semi-static scenery pieces were organized into distinct classes (Tree, Bush, Mountain, Lake, and Park), managing their own rendering coordinates and relative placement alongside the road boundaries. An overall Environment manager coordinates the spawning, positioning, and rendering order of these assets to maintain proper visual layering.

Adding Parallax Scrolling for Depth Implemented multi-layered parallax scrolling by updating background component positions at varying speeds relative to the player's movement. Distant elements like mountains move very slowly, mid-ground elements like trees and bushes move at a moderate pace, and foreground elements move fastest. This layered motion successfully simulates a strong three-dimensional sense of depth and speed.

Implementing Animated Elements and Dynamic Sky Developed dynamic atmospheric features, including animated `Cloud` objects drifting horizontally across the screen and `Bird`

sprites featuring frame-based flapping animations. Furthermore, a dynamic dawn-to-dusk sky system was integrated, gradually shifting background color gradients over time or distance traveled to simulate a realistic passage of day and night.

3. Deliverables

1. **Environment Classes:** Modular structure featuring `Tree`, `Bush`, `Mountain`, `Lake`, and `Park` components managed via an Environment controller.
2. **Parallax Scrolling System:** Multi-speed background layer movement creating depth perception.
3. **Animated Elements:** Animated `Bird` and `Cloud` objects for living atmospheric detail.
4. **Dynamic Sky:** Dawn-to-dusk transitional sky rendering integrated into the game panel loop.

4. Tools & Resources

- **Development Environment:** Java JDK 8+, IntelliJ IDEA / VS Code.
- **Reference Materials:** Java 2D Graphics rendering loops, custom multi-layered background scrolling algorithms, and state-based color gradient interpolation techniques for the dynamic sky.